

Types of Research Validity

1. Measurement 2. External 3. Internal 4. Statistical Conclusion

Research Hypotheses

(and evidence required to support each)

1. attributive
 - a way to measure the behavior
 - how to discriminate it from related behaviors
2. associative
 - reliable statistical relationship
3. causal
 - temporal precedence (cause before effect)
 - reliable statistical relationship
 - no confounds/alternative causal explanations

Kinds of Measures/Behaviors

	Variable	Constant
Measured/Subject		
Manipulated/Procedural		

Research Design

Which participants do what when?

Causal Interpretability	Between groups	Within-groups
True Exp.	BG True Exp.	WG True Exp.
Non Exp.	BG Non Exp.	WG Non Exp.

Stages of Participant Sampling

1. Target Population
2. Sampling Frame (complete pop or purposive)
3. Selected Sample (research selected or self-selected & simple or stratified)
4. Data sample (volunteerism & attrition)

Participant Sampling Decisions

(and choices)

1. Complete population or purposive sampling frame
2. Researcher selected or invited/self-selected
3. Simple or stratified

Data Collection

1. Collection – observation, self-report or trace
2. Setting -- laboratory, structured setting, or field
3. Data source – primary or archival

Types of Participant Assignment

1. RA of individuals by the researcher
2. RA of intact groups
3. Arbitrary
4. Administrative
5. Self-assignment
6. Non-assignment ("measured IV")

Internal Validity

(components and type of variables involved)

1. Initial Equivalence (measured/subject vars)
2. Ongoing Equivalence (manipulated/ procedural vars)

External Validity

(components and type of variables involved)

1. Population (measured/subject)
2. Setting (manipulated/procedural vars)
3. Task/Stimulus (manipulated/procedural vars)
4. Societal/Temporal

Role of each Measure/Behavior

1. Causal/Independent variable
2. Effect/Dependent variable
3. Control
 - variable
 - constant
4. Confounding variable